A Novel Coarse-to-Fine Deep Learning Registration Framework for Multimodal Remote Sensing Images

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***Abstract*— Multimodal remote sensing images with large rotation transformation (RT) are challenging to be registered. It needs to deal with the global geometric deformation caused by great RT and significant local appearance differences caused by different imaging mechanisms. Existing deep learning methods mainly use a single deep descriptor learning (DDL) network to extract invariant features for identifying matching samples and discriminative feature descriptors for separating nonmatching samples. However, it is difficult to extract local invariant feature descriptors to RT and modality change through a single DDL network. This article proposes a novel coarse-to-fine deep learning image registration framework for multimodal remote sensing images based on two task-specific deep models. Specifically, in the coarse registration stage, this article designs an effective deep ordinal regression (DOR) network for rotation correction, which can reduce the difficulty of multimodal image registration and boost image registration. The proposed DOR network transforms the rotation correction task into a rotation ordinal regression problem, which can exploit the potential relationship between the rotation ordinals to improve the accuracy of rotation estimation. In the fine registration stage, we adopt the DDL network to deal with the image modality change based on the rotation-corrected images. Extensive experimental results on multimodal image datasets demonstrate the significant advantages of the proposed coarse-to-fine deep learning registration framework. The DOR network achieves higher rotation correction accuracy, which can significantly improve the multimodal image registration performances.**

***Index Terms*— Coarse-to-fine, deep features, image registration, multimodal image, ordinal regression.**

I. Introduction

MULTIMODAL image registration aims to align images that are acquired from the same scene, but by different sensors [1], [2], [3]. Due to multimodal images containing complementary information, multimodal image registration is crucial in various remote sensing image applications, such as change detection [4], [5], [6], object detection [7], [8], image fusion [9], [10], [11], targets tracking [12], image classification [13], [14], and image segmentation [15]. However, there are significant differences between multimodal images, which increase the challenges of multimodal remote sensing image registration.

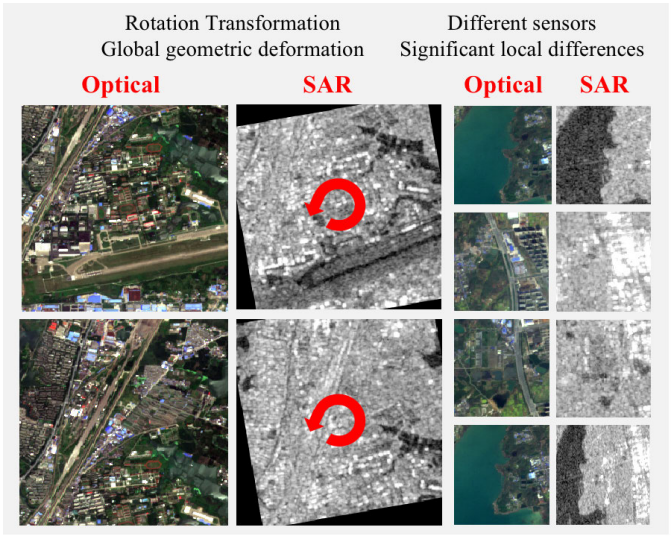


Fig. 1. Multimodal remote sensing images with large RT are challenging to be registered by a single DDL model, which should simultaneously deal with the global geometric deformation caused by RT and significant local appearance differences caused by various imaging mechanisms of different sensors.

Traditional image registration methods are mainly classified into intensity-based methods and feature-based methods. Intensity-based methods search for the optimal transformation parameters based on the image similarity of pixel intensity information, such as mutual information (MI) [16] and cross correlation (CC) [17]. However, image intensity is sensitive to image changes and affects registration accuracy. Traditional feature-based methods mainly use handcrafted descriptors to find the matching points and build the local correspondences between images. The handcrafted descriptors usually involve gradient statistic information or structure features [18], [19], [20], [21], [22], [23], [24]. The extracted local features are more robust to image changes than intensity, which are widely

Manuscript received 30 December 2022; revised 18 June 2023; accepted 2 August 2023. Date of publication 17 August 2023; date of current version 1 September 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62201407, Grant 62271377, and Grant 61977052; in part by the Key Research and Development Program of Shaanxi under Grant 2023-YBGY-244, Grant 2021ZDLGY0106, and Grant 2022ZDLGY0112; in part by the National Key Research and Development Program of China under Grant 2021ZD0110404; in part by the Key Scientific Technological Innovation Research Project by Ministry of Education; in part by the China Postdoctoral Science Foundation under Grant 2022M722496; in part by the Fundamental Research Funds for the Central Universities under Grant ZYTS23064; and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX23020. *(Corresponding author: Dou Quan.)*

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Digital Object Identifier 10.1109/TGRS.2023.3306042

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used for remote sensing image registration. However, handcrafted features are difficult to deal with the various image changes, such as illumination change, image modality change, and geometric transformation. To solve this problem, several works proposed a two-stage registration approach, which combines the intensity-based and feature-based methods [10], [25], [26] or employs multiple features [27], [28], [29] for coarse-to-fine image registration. In addition, hierarchical image registration based on multilevel image resolutions can achieve fast and accurate image registration results [30], [31], [32]. However, there are significant appearance differences caused by different imaging mechanisms in multimodal images. The handcrafted descriptor extraction method makes it hard to extract shared features between multimodal images and results in worse matching performances. Meanwhile, the traditional feature-based methods have a fixed feature extraction pipeline and lack adaptiveness for various images.

Recently, deep descriptors have gradually replaced handcrafted descriptors and have shown significant advantages in local image patch matching [33], [34], [35]. The data-driven learning model and multiple nonlinear mappings improve the adaptiveness of deep descriptors for multimodal images. Compared with handcrafted features, the DDL network can exploit stronger image modality-invariant information and more discriminative features for multimodal image matching. However, the performance of the DDL network will degenerate significantly when the multimodal images have a large rotation transformation (RT). Under this circumstance, the DDL network needs to deal with the global geometric deformation caused by RT and significant local differences in appearance caused by various imaging mechanisms at the same time, as shown in Fig. 1. It is challenging to learn invariant features to various image changes simultaneously by a single deep descriptor learning (DDL) model, which is not as good as learning each invariant feature with a specific model. In addition, we find that if the deep learning model blindly pursues the invariant features for various changes, it will affect the discrimination of features, degrading the performance of image matching.

Inspired by the traditional two-stage registration methods, this article proposes a novel coarse-to-fine deep learning registration framework for multimodal remote sensing images. We divide the multimodal image registration into two stages: the rotation correction coarse stage and the fine image registration stage. In the coarse registration stage, we propose the DOR network to estimate the rotation parameter and perform rotation correction by exploiting the global features of input images. Specifically, this article converts the rotation correction problem to an ordinal regression problem and carefully designs three effective DOR networks for rotation ordinal regression. The DOR network can be optimized by multiple binary cross-entropy losses. It can effectively exploit the relationship between rotation ordinals, facilitating precise rotation correction. In the fine registration stage, we use a DDL network to learn the local image modality-invariant features based on the rotation-corrected images. DDL network mainly deals with the significant appearance differences between multimodal images. It can be optimized by a triplet matching loss, which constrains the feature distance of the negative patch-pair to be larger than that of the positive patch-pair. Different from existing deep learning methods that merely use a single network to learn invariant features for various changes, this article uses two task-specific deep networks and optimization losses for multimodal image registration, which can significantly reduce the difficulty of multimodal image registration and improve their performance. In addition, it should be noted that DDL is effective in slight RT and can tolerate the small rotation correction error in the first phase. Thus, the proposed multimodal image registration method is robust.

The main contributions can be summarized as follows.

1) This article provides an insightful analysis of why the single DDL model makes it difficult to acquire accurate multimodal image registration. In addition, this article proposes a novel coarse-to-fine deep learning registration framework for multimodal remote sensing images, including the rotation correction stage based on deep ordinal regression (DOR) learning and the fine image registration stage based on DDL.
2) To the best of our knowledge, this article first transforms the image rotation correction task into an ordinal regression problem and designs three effective DOR networks for rotation ordinal regression. The DOR network can fully exploit the relationship between rotation ordinals and further boost the accuracy of rotation correction.
3) Extensive experimental results and analysis show the superiority of the DOR network on rotation correction and the effectiveness of coarse-to-fine multimodal image registration.

The rest of the article is organized as follows. Section II introduces the DDL methods and ordinal regression methods. Section III presents our proposed coarse-to-fine deep learning registration method. The experimental results and analysis are shown in Section IV. Section V draws a conclusion of this article.

II. Related Work

This part will briefly introduce image registration methods, discuss coarse-to-fine image registration, and present the related works of ordinal regression.

A. Image Registration Methods

Image registration methods mainly contain intensity-based methods and feature-based methods [2].

1) Intensity-Based Image Registration Methods: Intensity-based image registration methods find the optimal image transformation parameters by maximizing the image similarity based on pixel intensity [16], [17], [36]. Due to their similarity metrics being computed on raw image intensity information, intensity-based methods are sensitive to image changes and acquire desirable registration results under significant image appearance differences caused by image illumination changes and image modality changes.

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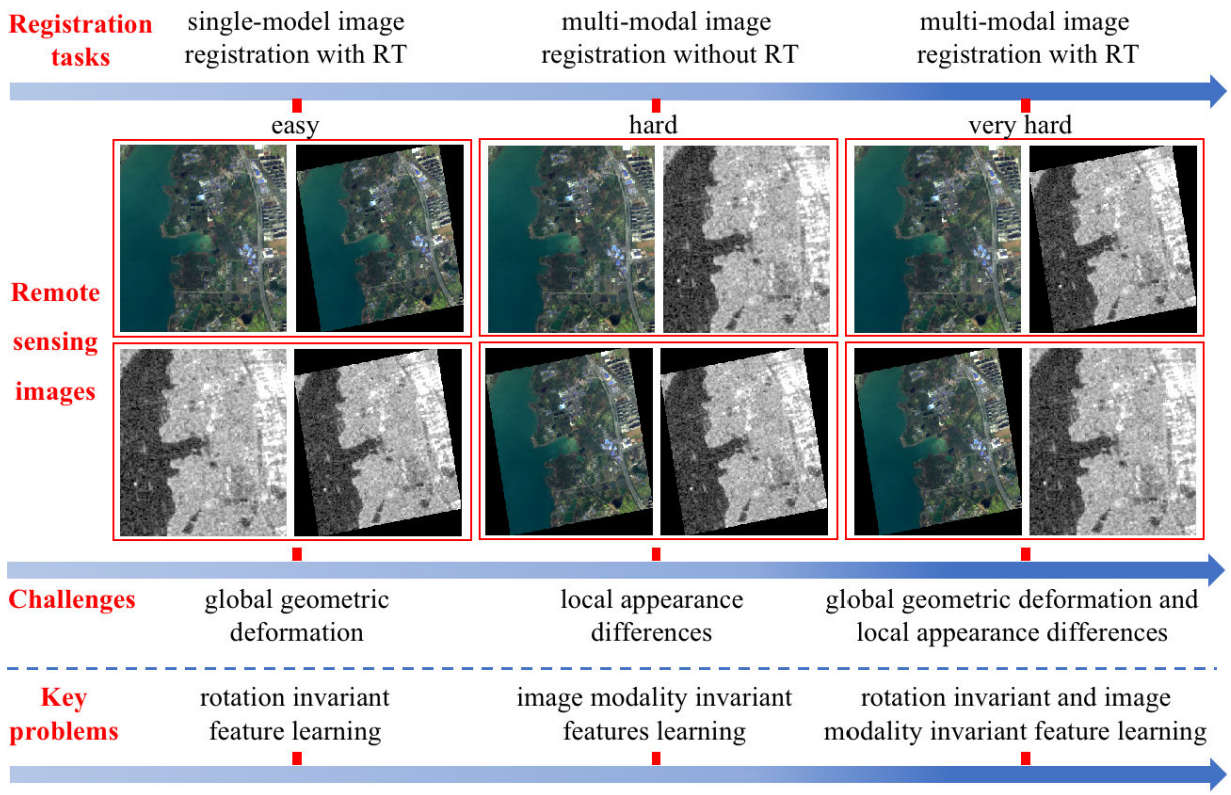


Fig. 2. Challenges and critical problems of feature-based methods in different remote sensing image registration tasks. The multimodal remote sensing images with global geometric deformation and local appearance differences are more difficult than single-modal image registration and multimodal image registration without RT.

2) Feature-Based Image Registration Methods: Image features are more robust to image changes than image intensity information, which are used for building local correspondences between images and registering images accurately [18], [37], [38]. The representative local features are statistical gradient features, texture features, shape features, and phase features. scale-invariant feature transform (SIFT) [18] and its various modified versions, speeded-up robust features (SURF) [19], Affine-SIFT [22], SAR-SIFT [21], and SIFT-OCT [20], have been widely applied for remote sensing image registration. To extract more robust descriptors for multimodal remote sensing image matching, [39] proposed a co-occurrence filter space matching method. Zhang et al. [40] proposed the histogram of the orientation of the weighted phase method, which can deal with the nonlinear radiation distortion and significant contrast differences between cross-modal images based on image phase features. However, the invariance and discrimination of traditional handcrafted features are insufficient to achieve stable and accurate registration performances on multimodal remote sensing images with significant appearance differences and serious speckle noise.

3) DDL Methods: With the development of deep learning, deep descriptors are applied for image matching and registration. DDL methods mainly map a local image patch to a feature vector for image matching [34], [35], [41], [42], [43], [44]. DDL networks (MatchNet [41], L2-Net [34], and HardNet [35]) are composed of several deep convolutional blocks that consist of convolutional layers, pooling operators, normalization layers, and nonlinear activation functions. Ideally, the learned matching descriptors should be similar, while the nonmatching descriptors should be dissimilar. DDL networks can be optimized by the contrastive loss [33], [44], the triplet loss [34], [35], and the ratio form of triplet loss (ratio loss) [44]. Specifically, the contrastive loss constrains the feature distance of matching patch-pairs to be as small as possible and that of nonmatching patch-pairs to be larger than a threshold. The triplet loss requires the feature distance of matching patch-pairs to be smaller than that of nonmatching patch-pairs by a margin. To further boost the descriptor learning, HardNet [35] proposed to exploit hard negative patch-pairs for network optimization. The hard negative patch-pairs have a small feature distance and contain rich guidance information for network training. Thus, HardNet achieves significant advantages in image matching and registration. To exploit rich information from feature similarity for boosting deep network optimization, [45] proposed a self-distillation feature learning network (SDNet) for optical and synthetic aperture radar (SAR) image registration. Wei et al. [46] proposed a deep modality independent descriptor learning network (DMID-Net) for matching, which computes the self-similarity of deep features as the shared local structure descriptor between multimodal images.

As shown in Fig. 2, the function of feature-based methods is to deal with global geometric deformation and extract rotation invariant features in single-modal image registration with RT. The key problem of multimodal image registration without RT is dealing with the significant local differences between images and learning image modality invariant features for matching.

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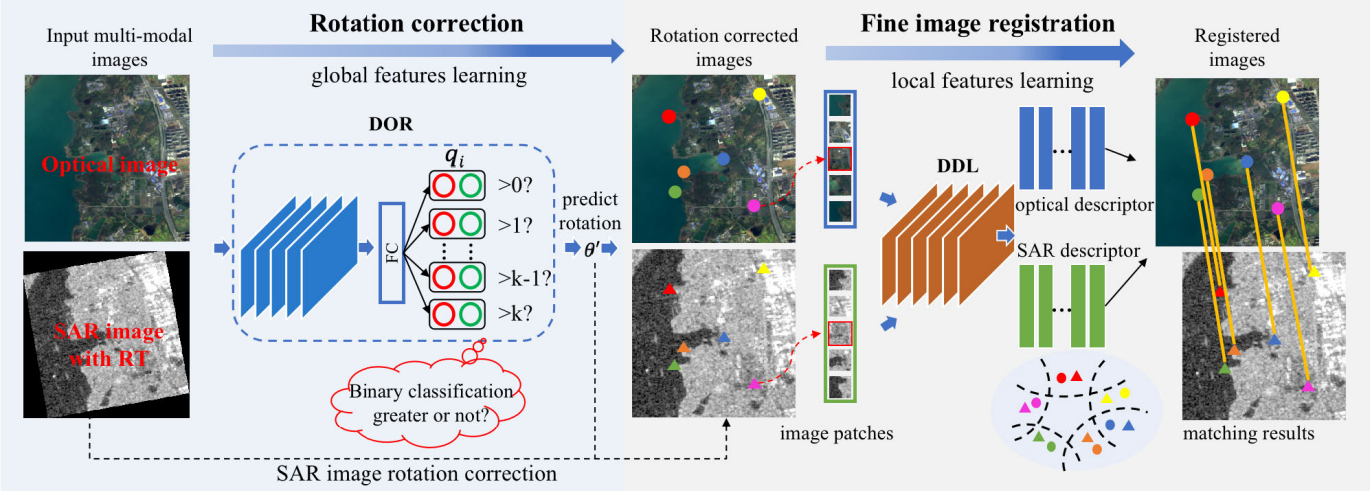


Fig. 3. Framework of our proposed DOR network-based multimodal image registration method, DOR-DDL. It contains the rotation correction stage and fine image registration stage. First, we propose a DOR network to lean global features for rotation parameter estimation and then perform rotation correction. Second, we use a DDL network to learn local image modality-invariant features for precise image registration.

However, multimodal image registration under the great RT is challenging. The main reason is that the single deep learning model makes it difficult to solve the great global geometric deformation and significant local differences between multimodal images at the same time. The DDL network should extract rotation invariant and image modality invariant features in multimodal remote sensing image registration with RT. Thus, to reduce the image registration difficulty and obtain accurate registration performances, this article proposes a novel coarse-to-fine deep learning framework that deals with rotation deformation and image modality change problems by two task-specific deep networks separately.

B. Discussion of Coarse-to-Fine Image Registration

To achieve more accurate and fast image registration, some works propose the two-stage coarse-to-fine image registration strategy that progressively achieves the optimal image registration result by combining image intensity information and features, multiple features, and multiresolution image features [10], [25], [26], [27], [28], [29], [47], [48], [49], [50]. These coarse-to-fine methods are mainly divided into three classes. First, they adopt local features for coarse registration and then conduct fine registration based on image similarity [10], [25], [26]. Second, they use a kind of image feature to achieve coarse registration results, and then adopt another type of feature to remove false matching points and obtain fine image registration performances [27], [28], [29]. In addition, the coarse-to-fine hierarchical image registration is performed on various image resolutions [30], [31], [32], [49], [51]. Generally, high-resolution image registration needs expensive computational costs. We can first conduct fast registration and obtain coarse registration results based on low-resolution images and then use them to guide the fine image registration based on high-resolution images.

The traditional two-stage registration method and the proposed method have similar motivations. It is difficult to achieve accurate registration performance through one stage or single model, and thus, image registration can be performed through several stages and multiple models. The main difference between the existing two-stage and our methods is that this article proposes a novel coarse-to-fine deep learning registration framework based on two task-specific networks. In the coarse stage, this article focuses on dealing with the RT by the DOR network. In the fine stage, this article mainly deals with the image modality changes and pursues fine registration performances by the DDL network.

C. Ordinal Regression

The ordinal regression problem [52] involves estimating an ordinal number, such as score, age, speed, height, or direction. There are many methods for solving the ordinal regression problem, such as the support vector learning approach [53] and a series of binary classification subtasks [54]. In each binary classification subtask, we need only to determine whether the ordinal number is greater than a value, which is much easier than directly estimating an exact ordinal number. Moreover, ordinal regression methods consider the relationship between ordinal numbers and acquire significant advantages in various applications, such as age estimation [55], motion speed estimation [56], credit rating [57], facial expression recognition [58], height estimation [59], and anomaly detection [60]. In this article, the rotation estimation task is naturally viewed as an ordinal regression problem. We first propose a DOR method for rotation correction in the multimodal image registration task. It can acquire a very small error in rotation estimation and achieve accurate rotation correction, further enhancing the performance of multimodal image registration.

III. Method

The framework of the proposed two-stage coarse-to-fine deep learning approach for multimodal image registration is shown in Fig. 3. It mainly contains two stages: the rotation

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correction stage and the fine image registration stage. First, we propose a DOR network to estimate the rotation angle between images and perform rotation correction for the input images. Second, the DDL network is responsible for subsequent fine image registration based on the rotation-corrected multimodal images. Thus, we named the proposed coarse-to-fine image registration method as DOR-DDL. In the training process, the DOR network is optimized by multiple binary cross-entropy losses, while the DDL network is optimized by the triplet matching loss. The details of the proposed image registration approach, network structures, and optimization methods are introduced as follows.

A. DOR Learning

In this section, we start by converting the rotation estimation problem into an ordinal regression problem, which can be implemented by a series of binary classification subtasks. Next, we design three effective networks for rotation ordinal regression and finally introduce the optimization loss for the DOR network.

Suppose that the multimodal image-pair for registration is $\{(x_i^1, x_i^2)\}_{i=1}^N$. To convert the rotation estimation problem into an ordinal regression task, we define the rotation angle between the input image-pair (x_i^1, x_i^2) as a discrete value θ_i, and $\theta_i \in [0, k]$. The goal of the DOR network is to estimate the rotation value between multimodal image-pairs. It can be implemented by a series of binary classification subtasks. The role of the jth binary classification subtask is to determine whether the rotation angle is larger than j. If $\theta_i > j$, the corresponding binary label of the jth classifier y_i^j is 1, otherwise y_i^j is 0, $j = 0 \ldots k$

$$y_i^j = \begin{cases} 1, & \text{if} \quad \theta_i > j \\ 0, & \text{if} \quad \theta_i \leq j. \end{cases} \tag{1}$$

The estimated rotation angle θ_i' is then determined by accumulating the predicted label $y_i^{j'}$ through the DOR network. θ_i' is formulated as

$$\theta_i' = \sum_{j=0}^{k} y_i^{j'}. \tag{2}$$

1) DOR Network: As shown in Fig. 4, we design three networks for rotation estimation, including a 2-channel network, a Siamese-vector network, and a Siamese-map network. They consist of several convolutional blocks. The main difference between these three networks is that they concatenate image information at different levels, such as the image-level, the middle-level, and the high-level.

In the 2-channel network, images are concatenated along the channel dimension and input to a deep convolution network, then output the predicted ordinal label $y_i^{j'}$ for each binary classification. Moreover, the Siamese-vector network uses two same deep convolutional networks to extract high-level feature vectors from the input images, respectively, and then concatenates them for binary label prediction. The Siamese-map network first extracts middle-level feature maps from input images, respectively, and concatenates them along the channel dimension for binary classification. In Section IV-D2, we will further compare the rotation estimation performances of these networks.

2) DOR Optimization: In the training process, the DOR network can be optimized by multiple binary cross-entropy losses. A binary classification subtask corresponds to a binary cross-entropy loss. The DOR loss L_{DOR} can be represented as

$$L_{\text{DOR}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=0}^{k} \left(y_i^j \log(q_i^j) + (1 - y_i^j) \log(1 - q_i^j) \right) \tag{3}$$

where q_i^j is the corresponding output probability of DOR network in the jth binary classifier.

3) Other Rotation Estimation Methods: Actually, there are other feasible rotation estimation methods. For example, we can directly regress the rotation parameter by a deep network, which is denoted as deep regression (DR). The optimization goal of DR is to minimize the estimation error between the output rotation parameter and the ground truth. The loss of DR L_{DR} can be represented as

$$L_{\text{DR}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \theta_i' - \theta_i \right\|. \tag{4}$$

In addition, the rotation estimation can also be implemented by the multiclass classification (MC) network. The rotation parameters from 0 to k can be classified into $k+1$ classes. The MC network can be optimized by minimizing the multiclass cross-entropy loss L_{MCE}

$$L_{\text{MCE}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=0}^{k} l_{ij} \log(p_{ij}) \tag{5}$$

where p_{ij} means the probability of the input image-pair (x_i^1, x_i^2) belongs to the j class and l_{ij} represents the corresponding one-hot label of the ground truth.

As previously mentioned, DOR, DR, and MC can be used for rotation estimation. Although the MC based on the deep convolutional network achieves significant performances, it ignores the relationship between different classes. Conversely, DOR can exploit this relationship to boost the rotation estimation. Specifically, the MC method merely expects the predicted rotation θ_i' to be consistent with the true rotation label θ_i. Inversely, DOR not only expects θ_i' to be consistent with θ_i but also encourages the estimated rotation to be close to the ground truth. Therefore, compared with the MC method, the DOR method will achieve a smaller rotation estimation error, which benefits to improving the image registration performances in the second stage. In addition, our proposal is more accessible to implement than directly predicting an exact rotation value in DR because DOR only needs to determine whether the rotation parameter is greater than a value in each binary classification subtask. Thus, DOR has significant advantages over DR in rotation estimation. In Section IV-D3, we will compare these rotation estimation methods for rotation estimation.

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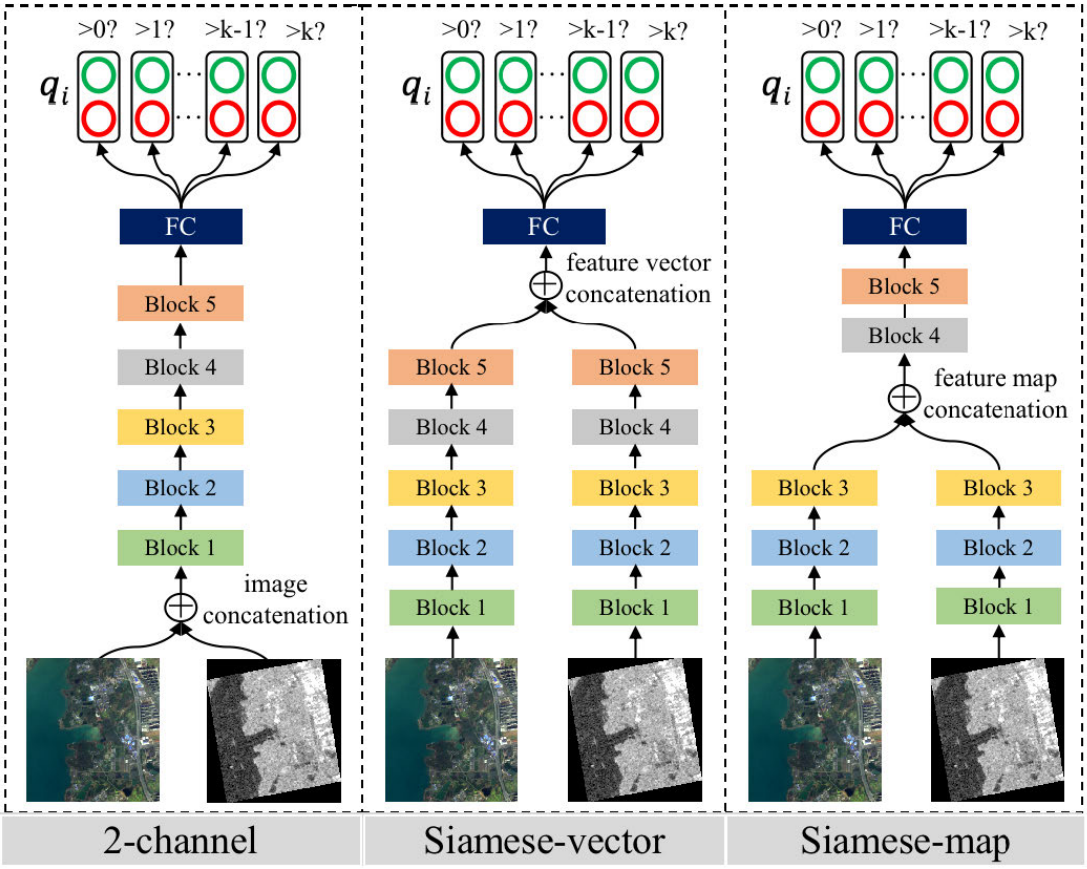


Fig. 4. Various network structures for rotation estimation.

B. Deep Descriptor Learning

According to the estimated rotation angle by the DOR network, we conduct rotation correction for the input image. After that, the rotation-corrected images are prepared for further fine alignment by DDL. Specifically, we first detect the keypoints in the rotation-corrected images by SIFT. Then, local image patches can be clipped around keypoints. This article adopts the representative DDL network, HardNet [35], to learn local features from image patches, which maps the image patch to an L2-normalized feature vector with length 128. The learned local features can be represented as follows:

$$\begin{cases} f_i^1 = F(x_{pi}^1) \\ f_i^2 = F(x_{pi}^2) \end{cases} \tag{6}$$

where $\{(x_{pi}^1, x_{pi}^2)\}_{i=1}^M$ represents the generated image patch-pair from the rotation corrected images, $F(\cdot)$ is the mapping function in DDL network, f_i^1 and f_i^2 are extracted features, f_i^1 and f_i^2 are matching, and f_i^1 and f_j^2 are nonmatching, $i \neq j$.

DDL network is optimized by the triplet matching loss L_m

$$L_m = [m + D(f_i^1, f_i^2) - D(f_i^1, f_j^2)]_+ \tag{7}$$

where $[x]_+ = \max(x, 0)$, $D(f_i^1, f_i^2)$ represents the feature distance between the f_i^1 and f_i^2, m is a distance margin.

Obviously, if the nonmatching pair has a large feature distance, the value of L_m tends to be 0, which lacks effective supervised information for DDL training. Thus, to exploit the informative samples, we adopt the hard negative samples mining strategy [35], [61] to select the negative pairs with a small feature distance for DDL optimization.

The triplet matching loss with the hard negative samples mining strategy can be represented as follows:

$$\begin{aligned} L_m &= [m + D(f_i^1, f_i^2) - D_{\text{hard},i}]_+ \\ D_{\text{hard},i} &= \min(D(f_i^1, f_j^2), D(f_j^1, f_i^2)) \\ i &\neq j, j = 1 \dots B \end{aligned} \tag{8}$$

where $D_{\text{hard},i}$ is the distance of the ith hard negative samples and B is the size of a mini-batch.

IV. Experiments

To present the effectiveness of the proposed coarse-to-fine deep learning registration method, we perform experiments on three multimodal datasets. First, this article shows the necessity of separately conducting rotation correction and fine multimodal image registration. Then, we present the superiority of the proposed DOR method on rotation correction. Finally, this article evaluates the performance of DOR-DDL with representative methods on multimodal image registration.

A. Data Introduction

The experiments are conducted on three public multimodal datasets: SEN1-2, a remote sensing image dataset, VEDAI, an aerial image dataset, and VIS-NIR, a natural image dataset.

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Fig. 5. (Left) Optical remote sensing images and (right) corresponding SAR images in the SEN1-2 dataset.



Fig. 6. (Left) VIS aerial images and (right) corresponding NIR aerial images in the VEDAI dataset.

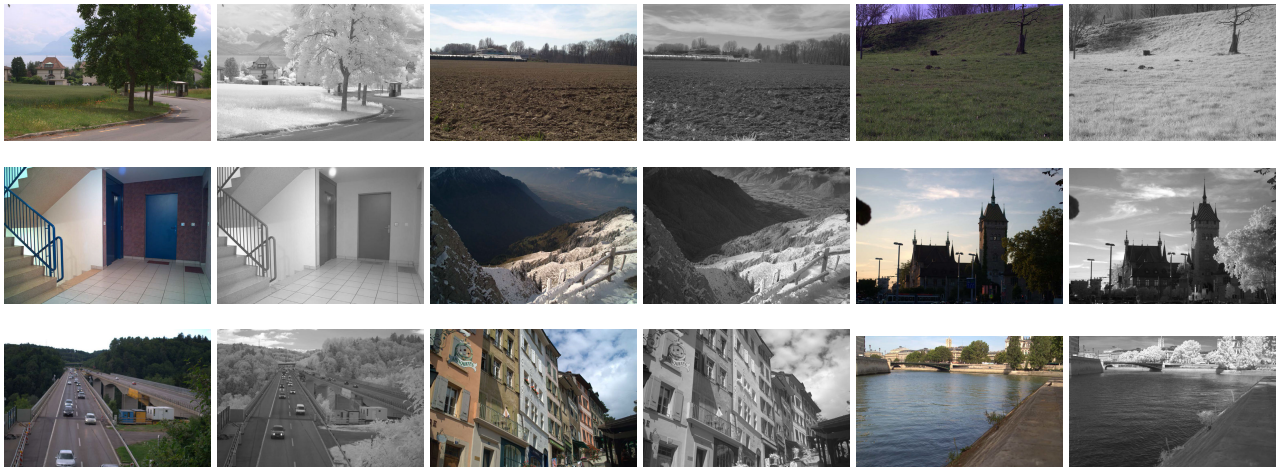


Fig. 7. (Left) VIS images and (right) corresponding NIR images in the VIS-NIR dataset. Image-pairs are acquired from nine scenes, including Country, Field, Forest, Indoor, Mountain, Oldbuilding, Street, Urban, and Water. There are obvious image content differences between different scenes.

SEN1-2 [62], [63], [64], [65] contains an optical remote sensing image and a corresponding SAR image with the size of 256×256. The image resolution is 10 m. To train the DOR network for rotation estimation, we generate 24 K optical-SAR image-pairs with rotation labels for training and 10 K optical-SAR image-pairs for testing. In addition, to train the DDL network, we build 583 K local image patch-pairs for training and 248 K image patch-pairs for testing. The size of the image patches is 64×64. Finally, 570 optical-SAR image-pairs are selected to test the performance of the multimodal image registration. As shown in Fig. 5, there are significant differences between optical and SAR images in the SEN1-2 dataset. The main challenges of the SEN1-2 dataset for registration are significant nonlinear differences between cross-modal images and the speckle noises in SAR images.

VEDAI [66], [67] is a benchmark dataset for vehicle detection in aerial imagery. It includes 1246 visible spectrum (VIS) and near-infrared (NIR) image-pairs with a size 512×512, in which 800 image-pairs are for training and 446 image-pairs for testing. To train the DOR network, we generate 25 600 training image-pairs and 10 704 test image-pairs by performing random clip, random flip, resize, and random rotation methods based on the original image-pairs. In addition, there are 140 K local image patch-pairs clipped from training image-pairs for DDL training. Figure 6 presents three VIS-NIR image-pairs in the VEDAI dataset. The VEDAI dataset shows slight differences between VIS and NIR images. However, the test images lack texture features in most local regions. Thus, it is not easy to extract discriminative features for image matching and registration.

VIS-NIR [68], [69], [70] is a cross-spectral natural dataset containing VIS and NIR images from nine scenes: Country, Field, Forest, Indoor, Mountain, Oldbuilding, Street, Urban, and Water, as shown in Fig. 7. The number of image-pairs are 52, 51, 53, 56, 55, 51, 50, 58, 51, respectively. First, we use random clip, resize, random flip, and random rotation methods to generate 17 720 VIS-NIR image pairs from the Country scene for rotation estimation training and construct 306, 318, 336, 330, 306, 300, 348, and 306 image-pairs from the other eight scenes, respectively, for testing. In addition, there are 138 K local image patch-pairs are acquired from the Country scene for DDL training. Finally, we perform image

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registration on the other eight scenes. The VIS-NIR dataset contains nine scenes with obvious image content differences. The cross-modal image registration model should have a strong generalization ability from the Country scene to other test scenes.

B. Implement Details

All of the experiments were performed on the GeForce RTX 2080 Ti. The image size for rotation estimation and registration was 256×256, and the size of local image patches was 64×64. In the training process, we adopted the Adam [71] as an optimizer. Specifically, the initial learning rate was 0.5 on SEN1-2, 5 on VEDAI, and 1 on VIS-NIR when optimizing the DOR network. The batch size was 64, and the training epoch was 10. In the training process of DDL, the initial learning rate was 0.5 on SEN1-2 and VIS-NIR, the initial learning rate was 5e-6 on VEDAI, the batch size was 500, and the training epoch was 10. The margin in triplet loss L_m was $m = 1$.

C. Evaluation Metric

This article uses FPR95 and matching accuracy (Acc_P) to evaluate the local image patch matching performance. FPR95 metric means the false positive rate when the true positive rate is 95%. In addition, we take the accuracy of rotation estimation (Acc_R) and mean square error of rotation estimation (MSE_R) to reflect rotation correction performances. As the output of DR is not an exact value, we use the rounded rotation to compute Acc_R. It can be represented as

$$\text{Acc}_R = \frac{\text{Num}_\theta}{\text{Num}_{\text{all}}} \tag{9}$$

where Num_θ is the number of image-pairs that are accurately rotation corrected, Num_{all} is the number of test image-pairs.

MSE_R can be computed as

$$\text{MSE}_R = \frac{1}{\text{Num}_{\text{all}}} \sum_{i=1}^{\text{Num}_{\text{all}}} \sqrt{\left\|\theta_i - \theta_i'\right\|^2}. \tag{10}$$

Moreover, we adopt the widely used metrics to evaluate the image registration performance, such as root mean square error (RMSE), the number of registered images (Num_m), the matching probability of images (I_{mp}), and the average number of acquired point-pairs (Num_p). If the RMSE is smaller than five and the images are visually aligned, the corresponding multimodal images can be seen as successfully registered.

D. Ablation Study

This part first investigates the necessity of the coarse-to-fine deep learning registration method. Then, we will test the effectiveness of the network structure and the DOR method for rotation correction.

1) Necessity of the Coarse-to-Fine Method: To show the necessity of the coarse-to-fine image registration method, we presented the local matching results of the single image change learning model (RT or image modality change) and the various image changes learning model (RT and image modality change). We constructed four sets of training and test datasets of image patch-pairs from SEN1-2, including single modal image patch-pairs with RT (Optical-RT, SAR-RT), multimodal image patch-pairs without RT (Optical-SAR), and multimodal image patch-pairs with RT (Optical-SAR-RT). If not specified, $\theta = 15$ in this part. The experimental results based on the representative DDL network, HardNet, are shown in Table I, and the training process is shown in Fig. 8.

First, from Acc_p and FPR95 in Table I, we can see that the matching results of multimodal images with RT are worse than single modal images with RT and multimodal images without RT. Specifically, the matching accuracy of Optical-SAR (94.62%) is larger than that of Optical-SAR-RT (92.43%). The FPR95 result of Optical-SAR (5.89) is far smaller than that of Optical-SAR-RT (11.64). In addition, Fig. 8(a) shows that the loss of Optical-SAR-RT is the largest. Thus, it is challenging to deal with the image modality change and RT at the same time.

Noticeably, when a single model is trained to deal with RT and image modality change simultaneously, matching results on the single change dataset will be degraded. Specifically, the Acc_p on Optical-SAR decreases from 94.62% to 88.67%, and the FPR95 on Optical-SAR increases from 5.89 to 23.24 when DDL is trained on Optical-SAR-RT. The main reason is that DDL is difficult to learn the rotation-invariant features and image modality-invariant features from local image patches simultaneously. The hybrid learning of various invariant features will significantly decrease the single invariance of features. In addition, to close the distance of matching multimodal images in the feature space, DDL methods blindly pursue image modality-invariant feature learning to overcome their significant appearance differences, which will lose a lot of discriminant information and degrade the matching performances of multimodal images. Thus, it is necessary to adopt the two-stage method for separately conducting the rotation correction and the fine image registration.

2) Effectiveness of the Network Structure: In Tables II, III, and IV, we investigate rotation correction performances of the designed three networks: 2-channel network, Siamese-vector network, and Siamese-map network. In the DOR method, three networks all achieve high accuracy of rotation estimation, which demonstrates the effectiveness of the proposed networks. Furthermore, the Siamese-map network shows significant advantages on SEN1-2. For example, Siamese-map network respectively increases the accuracy of 2-channel by 2.16%, 2.62%, and 4.98% based on DOR, MC, and DR. Siamese-map network also improves the accuracy of Siamese-vector network by 2.64%, 2.02%, and 5.37% based on three methods. The reason behind this phenomenon is that modality-specific features generally exist in shallow layers of the network, and the Siamese-map network can independently extract the modality-specific features at first and then conduct full feature fusion in the middle-level. Considering the superiority of the Siamese-map network, we use it in the following experiments.

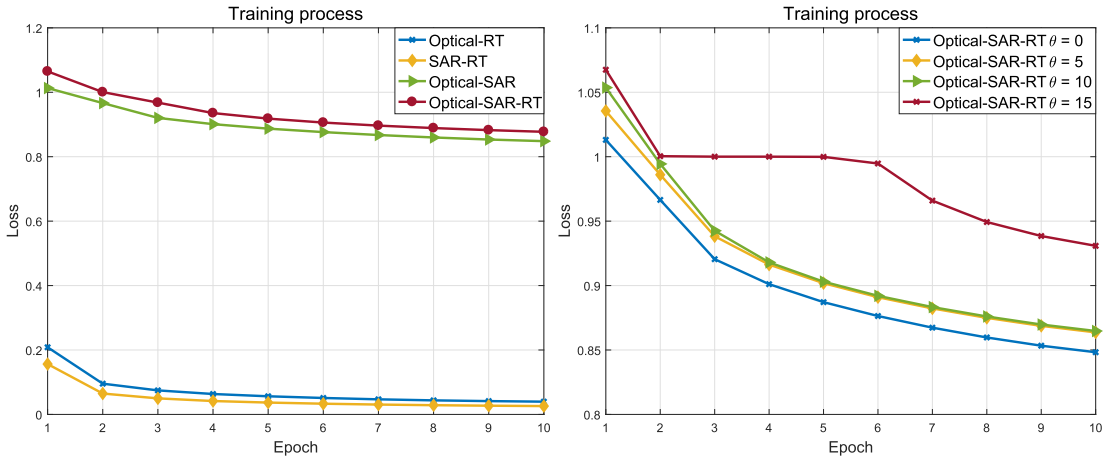
3) Effectiveness of DOR: To verify the effectiveness of DOR, Tables II, III, and IV compare the results of DOR with other rotation estimation methods, MC and DR. In terms of Acc_R, DOR shows slight advantages over MC on SEN1-2

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TABLE I

IMAGE PATCH MATCHING RESULTS (FPR95/ACC$_p$%) OF HARDNET ON DIFFERENT DATASETS

	Training data			
Testing data	Optical-RT	SAR-RT	Optical-SAR	Optical-SAR-RT
Optical-RT	**0.00/100**	**0.00/100**	34.90/86.00	0.05/99.53
SAR-RT	**0.00/100**	**0.00/100**	68.15/73.76	0.11/99.11
Optical-SAR	99.80/51.32	99.40/51.05	**5.89/94.62**	23.24/88.67
Optical-SAR-RT	99.80/51.37	99.51/51.28	68.00/73.76	**11.64/92.43**



(a) (b)

Fig. 8. Training process of HardNet on different datasets. (a) Different image registration problems. (b) Different RT.

TABLE II

ACC$_R$ AND MSE$_R$ OF ROTATION CORRECTION IN SEN1-2 IMAGE-PAIRS BASED ON DIFFERENT ROTATION CORRECTION METHODS

	Network structures & Rotation correction methods								
	2-channel			Siamese-vector			Siamese-map		
	DOR	MC	DR	DOR	MC	DR	DOR	MC	DR
Acc_R%	**95.64**	94.41	91.02	**95.16**	95.01	90.63	**97.80**	97.03	96.00
MSE_R	**0.1035**	0.1652	0.2904	**0.0718**	0.1080	0.2871	**0.0443**	0.0639	0.0983

TABLE III

ACC$_R$ AND MSE$_R$ OF ROTATION CORRECTION IN VEDAI IMAGE-PAIRS BASED ON DIFFERENT ROTATION CORRECTION METHODS

	Network structures & Rotation correction methods								
	2-channel			Siamese-vector			Siamese-map		
	DOR	MC	DR	DOR	MC	DR	DOR	MC	DR
Acc_R%	**98.62**	97.84	69.54	**98.48**	96.23	75.97	**98.48**	98.12	83.20
MSE_R	**0.0342**	0.0449	0.6063	**0.0531**	0.1168	0.4798	**0.0311**	0.0516	0.4873

and VEDAI. DOR presents clear superiority over DR on all test datasets. Specifically, DOR improves the accuracy of MC on SEN1-2 by 1.23%, 0.15%, and 0.77% based on the three networks. Moreover, DOR dramatically improves the accuracy of DR on SEN1-2 by 4.62%, 4.53%, and 1.8%. In addition, when the training dataset is small, DOR is superior to the MC and DR methods. For example, DOR improves the Acc$_R$ of MC by 2.99%, 4.51%, and 2.24%, and improves the Acc$_R$ of DR on VIS-NIR by 21.33%, 20.89%, and 25.82%. Moreover, DOR shows distinct advantages over MC and DR in terms of rotation estimation error. A small error can significantly contribute to the precise image registration. Thus, compared with MC and DR, DOR is clearly superior in terms of rotation correction. The superiority of DOR mainly comes from the two aspects. First, DOR can capture the relationship between rotations to boost the performance of rotation estimation. Second, the binary classification subtasks in DOR are easily optimized than the direct regression in DR. Therefore, DOR presents obvious advantages over MC and DR.

E. Comparative Results on SEN1-2

This section compares the optical-SAR image registration results of DOR-DDL with the traditional methods of SIFT [18], ORB [23], and PSO-SIFT [24]; the representative DDL methods of MatchNet with the contrastive loss (MatchNet-Contra), with triplet loss (MatchNet-Triplet), and with ratio loss (MatchNet-Ratio) [41], [44]; and HardNet [35] with the exponent loss (Exp-HardNet) [42], DMID-Net [46], and SD-Net [45]. The contrastive loss L_{con}, triplet loss L_m, and ratio loss L_{ratio} can be represented as follows:

$$L_{\text{con}} = \delta_+ + [\mu - \delta_-]_+$$
$$L_m = [m + \delta_+ - \delta_-]_+$$

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TABLE IV

ACC$_R$ AND MSE$_R$ OF ROTATION CORRECTION IN VIS-NIR IMAGE-PAIRS BASED ON DIFFERENT ROTATION CORRECTION METHODS

	Network structures & Rotation correction methods								
	2-channel			Siamese-vector			Siamese-map		
	DOR	MC	DR	DOR	MC	DR	DOR	MC	DR
	$Acc_R\%$								
Field	**94.77**	93.46	78.43	**92.48**	89.87	78.76	**96.73**	94.12	74.18
Forest	**98.74**	97.80	85.22	**98.11**	96.23	87.11	**99.06**	97.80	87.11
Indoor	**89.29**	86.01	58.63	**91.67**	86.31	61.31	**89.88**	89.29	53.57
Mountain	**96.06**	93.03	76.67	**97.27**	92.42	78.48	**95.15**	93.64	74.85
Oldbuilding	**87.25**	79.74	61.44	**90.52**	81.37	55.23	**86.27**	82.68	56.21
Street	**94.00**	92.33	74.67	**95.33**	91.00	76.33	**94.00**	91.00	72.00
Urban	**90.52**	85.92	63.22	**88.22**	81.32	64.66	**89.08**	87.07	52.87
Water	**89.54**	87.91	71.24	**87.91**	86.93	72.55	**90.20**	86.93	63.07
Mean	**92.52**	89.53	71.19	**92.69**	88.18	71.80	**92.55**	90.31	66.73
	MSE_R								
Field	**0.2092**	0.7026	1.0784	**0.7320**	0.7582	1.2810	**0.6699**	0.7157	1.4346
Forest	**0.0126**	0.0314	0.3459	**0.0189**	0.0472	0.2421	**0.0094**	0.0220	0.4528
Indoor	**0.8423**	1.116	1.3036	0.7530	**0.6220**	2.1637	**0.8482**	1.4792	2.0893
Mountain	0.1121	**0.0697**	0.5394	**0.0273**	0.0758	0.6030	0.2394	**0.0818**	0.8061
Oldbuilding	**0.3497**	0.6242	1.9641	**0.3105**	0.4771	3.1373	**0.2353**	0.6242	3.1438
Street	**0.6100**	1.6367	1.7400	**0.9533**	2.2633	2.8900	**0.8833**	1.0100	2.2867
Urban	**0.1782**	0.2787	1.0603	**0.2328**	0.3247	1.2356	**0.1983**	0.2529	1.7299
Water	**0.5719**	0.8301	1.1634	**0.5719**	0.6275	1.9869	**0.6699**	0.7647	2.3170
Mean	**0.3607**	0.6612	1.1494	**0.4499**	0.6495	1.6925	**0.4692**	0.6188	1.7825

$$L_{\text{ratio}} = \left(\frac{e^{\delta_+}}{(e^{\delta_+} + e^{\delta_-})}\right)^2 + \left(1 - \left(\frac{e^{\delta_-}}{(e^{\delta_+} + e^{\delta_-})}\right)\right)^2 \quad (11)$$

where δ_+ and δ_- are feature distances of positive and negative samples, respectively.

Table V shows the image registration results based on different degrees of RT, such as slight RT ($\theta \leq 5$) and large RT ($\theta \leq 10$ and $\theta \leq 15$). Traditional methods have difficulty dealing with optical-SAR images because there are significant differences in appearance. Although MatchNet can register some optical-SAR images, the matching probability needs to be further improved. DDL networks extract the shared features from multimodal images through several nonlinear convolutional layers. They can deal well with multimodal images with slight translation or small RT. DMID-Net and SD-Net mainly focus on learning image modality-invariant features, which acquire better performance than HardNet on multimodal images without RT. However, DMID-Net and SD-Net show slight advantages over DOR-DDL in the case of multimodal image-pairs with small RT. The main reason is that the DOR-DDL designs two task-specific deep networks for dealing with the RT and image modality changes, respectively. The rotation correction in the first stage will bring cumulative error into the second stage and decrease the registration accuracy, especially when the multimodal images are without RT. Most of the training samples for DOR network optimization have RT. DOR network inevitably introduces slight rotation estimation errors for the multimodal images merely with translation or small RT. In addition, DMID-Net and SD-Net cannot deal with the introduced rotation error, which is better at registering images without RT. Since HardNet can tolerate the small RT and is not affected by the slight cumulative error, DOR-DDL achieves similar results to HardNet.

When there are translation and large RT between multimodal images, DOR-DDL shows significant advantages over DDL networks, including HardNet, Exp-HardNet, DMID-Net, and SD-Net. As θ increases, the performance of DDL networks decreases sharply, while the image registration performance of DOR-DDL is relatively stable. The main reason is that it is very difficult to simultaneously deal with RT and image modality change for DDL. DOR can significantly reduce the difficulty of multimodal image registration by rotation correction. Specifically, when $\theta \leq 10$, DOR-DDL improves the image matching probability of HardNet by 5.79% and improves that of Exp-HardNet by 10.53%. When $\theta \leq 15$, DOR-DDL improves the image matching probability of HardNet by 22.98% and improves that of Exp-HardNet up 19.65%. Furthermore, DOR-DDL achieves a larger Num$_p$ and smaller RMSE than HardNet and Exp-HardNet. When $\theta \leq 10$, DOR-DDL improves the image matching probability of DMID-Net by 4.38% and improves that of SD-Net by 3.69%. When $\theta \leq 15$, DOR-DDL improves the image matching probability of DMID-Net by 11.57% and improves that of SD-Net by 14.04%. Thus, DOR-DDL is more effective in multimodal image registration with a large RT. The larger the RT, the more obvious the advantages of the DOR-DDL. In addition, these experimental results also demonstrated that the proposed coarse-to-fine deep learning registration framework could be flexibly combined with existing registration methods and improve their image registration performances.

F. Comparative Results on VEDAI and VIS-NIR

Figure 9 shows the image matching probability I_{mp} on VEDAI and each test subset of VIS-NIR. With the rotation increases, the results of DDL methods (MatchNet and HardNet) decline quickly. However, DOR-DDL still achieves stable results on the great RT. These results demonstrate the effectiveness of rotation correction based on DOR, which can reduce the difficulty of multimodal image registration. Furthermore, it should be noted that DDL methods are effective in slight RT ($\theta \leq 5$). They can tolerate the small rotation correction error from the first stage. Thus, our two-stage coarse-to-fine regression method DOR-DDL is quite stable and robust to RT.

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TABLE V

IMAGE REGISTRATION RESULTS ON SEN1-2. DOR-DDL (HARDNET, EXP-HARDNET, DMID-NET, AND SD-NET) MEANS THE LOCAL DESCRIPTOR LEARNING IS BASED ON HARDNET, EXP-HARDNET, DMID-NET, AND SD-NET, RESPECTIVELY

Method	Num_m	I_{mp}	$RMSE$	Num_p
ORB [23]	-	-	-	-
SIFT [18]	-	-	-	-
PSO-SIFT [24]	-	-	-	-
Translation and Rotation $\theta \leq 5$				
MatchNet-Contra [41], [44]	297	52.11%	1.43	497
MatchNet-Triplet [41], [44]	286	50.18%	1.42	502
MatchNet-Ratio [41], [44]	293	51.40%	1.41	505
HardNet [35]	**447**	**78.42%**	**1.31**	1000
DOR-DDL(HardNet)	**447**	**78.42%**	**1.31**	**1003**↑3
Exp-HardNet [42]	404	70.88%	1.33	762
DOR-DDL(Exp-HardNet)	**415**↑11	**72.81%**↑1.93%	**1.31**↓0.02	**779**↑17
DMID-Net [46]	415	72.81%	1.30	**817**
DOR-DDL(DMID-Net)	395↓20	69.30%↓3.51%	1.30	752↓65
SD-Net [45]	441	77.37%	1.26	**873**
DOR-DDL(SD-Net)	422↓19	74.04%↓3.33%	1.26	843↓30
Translation and Rotation $\theta \leq 10$				
MatchNet-Contra [41], [44]	268	47.02%	1.43	442
MatchNet-Triplet [41], [44]	257	45.09%	1.67	445
MatchNet-Ratio [41], [44]	271	47.54%	1.64	432
HardNet [35]	408	71.58%	1.36	838
DOR-DDL(HardNet)	**441**↑33	**77.37%**↑5.79%	**1.30**↓0.06	**966**↑128
Exp-HardNet [42]	351	61.58%	1.36	650
DOR-DDL(Exp-HardNet)	**411**↑60	**72.11%**↑10.53%	**1.31**↓0.05	**772**↑122
DMID-Net [46]	373	65.44%	1.34	**695**
DOR-DDL(DMID-Net)	**398**↑25	**69.82%**↑4.38%	**1.32**↓0.02	690↓5
SD-Net [45]	398	69.82%	1.33	736
DOR-DDL(SD-Net)	**419**↑21	**73.51%**↑3.69%	**1.30**↓0.03	**784**↑48
Translation and Rotation $\theta \leq 15$				
MatchNet-Contra [41], [44]	210	36.84%	1.47	426
MatchNet-Triplet [41], [44]	225	39.47%	1.45	407
MatchNet-Ratio [41], [44]	210	36.84%	1.43	424
HardNet [35]	303	53.16%	1.37	784
DOR-DDL(HardNet)	**434**↑131	**76.14%**↑22.98%	**1.31**↓0.06	**989**↑205
Exp-HardNet [42]	289	50.70%	1.36	592
DOR-DDL(Exp-HardNet)	**401**↑112	**70.35%**↑19.65%	**1.32**↓0.04	**770**↑178
DMID-Net [46]	310	54.39%	1.37	625
DOR-DDL(DMID-Net)	**376**↑66	**65.96%**↑11.57%	**1.32**↓0.05	**702**↑77
SD-Net[45]	326	57.19	1.36	675
DOR-DDL(SD-Net)	**406**↑80	**71.23%**↑14.04%	**1.31**↓0.05	**782**↑107

TABLE VI

FLOPS AND PARAMETERS OF DIFFERENT DOR NETWORKS

Networks	Input size	Parameters	FLOPs
2-channel-DOR	$2 \times 256 \times 256$	2.23M	0.17GMac
Siamese-vector-DOR	$2 \times 256 \times 256$	1.19M	0.29GMac
Siamese-map-DOR	$2 \times 256 \times 256$	2.36M	0.29GMac

G. Computation Cost

In the experiments, the average time of the whole image registration is 0.77 s for an image-pair, while the average time of the rotation correction stage is 0.09 s. Therefore, DOR does not significantly increase the time cost. That is to say, compared with the single-stage image registration method DDL, the two-stage coarse-to-fine deep learning registration method DOR-DDL increases a very small cost in time. In addition, Table VI shows that DOR is an efficient way to achieve high accuracy gains on multimodal image registration with a small amount of computation and an acceptable number of parameters.

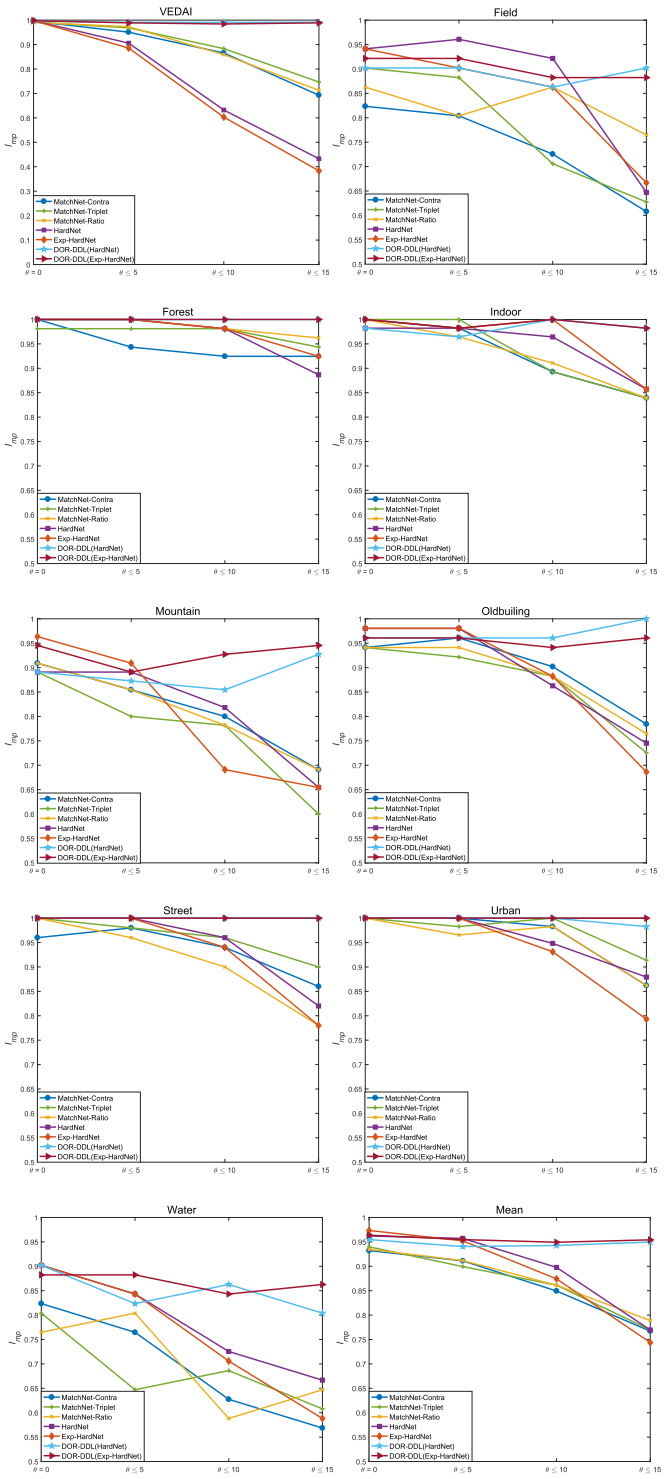


Fig. 9. Image matching probability I_{mp} of different methods on VEDAI and each test category of VIS-NIR.

H. Larger RT

To further verify the effectiveness of our proposal, we present the rotation correction results on multimodal images with larger RT, $\theta \leq 30$.

Tables VII, VIII, and IX show the rotation estimation accuracy and rotation estimation error of SEN1-2, VEDAI, and VIS-NIR. When $\theta \leq 30$, the proposed DOR method

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TABLE VII

ACC$_R$ AND MSE$_R$ RESULTS OF ROTATION CORRECTION IN THE SEN1-2 DATASET, WHERE $\theta \leq 30$

	Network structures & Rotation correction methods								
	2-channel			Siamese-vector			Siamese-map		
	DOR	MC	DR	DOR	MC	DR	DOR	MC	DR
$Acc_R\%$	**94.69**	94.55	90.10	**94.32**	88.67	92.07	**96.90**	96.96	93.19
MSE_R	**0.2456**	0.3302	0.6190	**0.1537**	0.7171	0.4145	**0.0843**	0.1643	0.1963

TABLE VIII

ACC$_R$ AND MSE$_R$ RESULTS OF ROTATION CORRECTION IN THE VEDAI DATASET, WHERE $\theta \leq 30$

	Network structures & Rotation correction methods								
	2-channel			Siamese-vector			Siamese-map		
	DOR	MC	DR	DOR	MC	DR	DOR	MC	DR
$Acc_R\%$	**98.44**	97.60	86.07	**97.90**	93.59	70.49	**98.27**	97.60	70.81
MSE_R	**0.0817**	0.3404	0.5353	**0.1470**	0.6921	0.8545	**0.0818**	0.1129	0.9697

TABLE IX

ACC$_R$ AND MSE$_R$ RESULTS OF ROTATION CORRECTION IN THE VIS-NIR DATASET, WHERE $\theta \leq 30$

	Field	Forest	Indoor	Mountain	Oldbuilding	Street	Urban	Water	Mean
				$Acc_R\%$					
2-channel DOR	**93.46**	**97.48**	**91.07**	**98.18**	**89.54**	**95.67**	**86.78**	**92.48**	**93.08**
2-channel MC	91.18	97.17	86.90	93.33	83.01	90.33	85.92	90.20	89.75
2-channel DR	76.47	82.70	59.52	70.30	61.11	71.67	58.91	71.24	68.99
Siamese-vector DOR	**93.46**	**95.60**	**91.37**	**94.55**	**89.87**	**94.00**	**87.07**	**91.18**	**92.14**
Siamese-vector MC	81.70	87.11	80.06	87.27	75.16	83.33	72.41	80.39	80.93
Siamese-vector DR	26.14	25.16	14.58	21.52	16.34	25.00	20.98	21.24	21.37
Siamese-map DOR	**96.73**	**98.74**	**91.96**	**96.36**	**92.48**	**94.67**	**88.22**	**89.54**	**93.59**
Siamese-map MC	88.89	95.28	82.74	90.91	79.41	91.00	81.90	85.95	87.01
Siamese-map DR	57.84	72.33	45.83	58.79	45.75	61.67	49.71	56.54	56.06
				MSE_R					
2-channel DOR	**0.0654**	**0.0252**	**0.4732**	0.2303	**0.1340**	**0.0633**	**0.5833**	**4.6373**	**0.7765**
2-channel MC	0.1078	0.0377	4.3810	**0.0667**	2.8072	0.1733	0.8247	13.9085	2.7884
2-channel DR	1.1242	2.5629	2.0179	1.2000	6.2810	2.1833	4.7414	6.8399	3.3688
Siamese-vector DOR	**0.0654**	**0.0440**	**0.0863**	**0.3970**	**0.1209**	**0.0867**	**0.7874**	0.8758	**0.3079**
Siamese-vector MC	2.6176	1.0440	1.3065	0.9788	0.3529	0.9533	4.2730	**0.2745**	1.4751
Siamese-vector DR	3.5948	5.0000	17.3929	6.6394	9.4641	5.4567	11.9080	5.9379	8.1742
Siamese-map DOR	**0.0425**	**0.0220**	**0.6012**	**0.0455**	**0.2353**	**0.1167**	0.3362	**1.5261**	**0.3657**
Siamese-map MC	0.1405	0.0472	4.6280	0.0909	0.3268	0.1500	**0.2213**	14.0882	2.4616
Siamese-map DR	2.0294	2.0975	3.1875	3.7303	5.6993	1.6867	5.9943	3.5131	3.4923

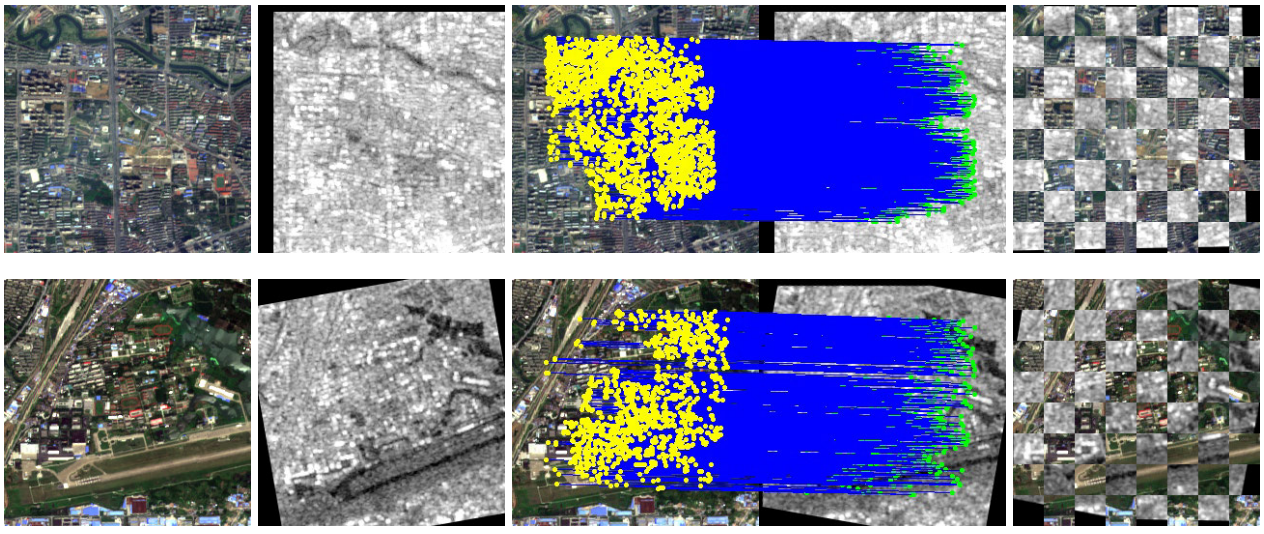


Fig. 10. Multimodal image registration results of the randomly selected optical-SAR images in the SEN1-2 dataset. (Left to right) Optical image, SAR image, matching point-pairs, and checkerboard image.

still achieves higher accuracy and smaller error than MC and DR methods. Especially on VIS-NIR, DOR shows significant advantages over the other two methods. These results demonstrate the effectiveness and robustness of DOR for rotation correction under the large RT. Specifically, the MSE$_R$ of DOR on all test datasets are smaller than 1, which will reduce the

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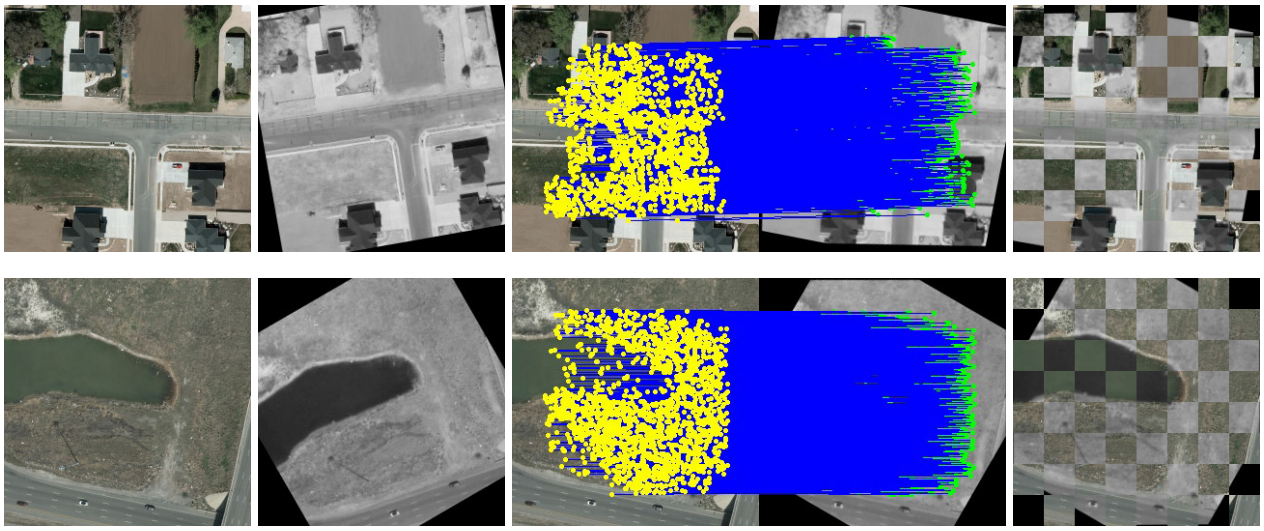


Fig. 11. Multimodal image registration results of the randomly selected VIS-NIR images in the VEDAI dataset. (Left to right) VIS image, NIR image, matching point-pairs, and checkerboard image.

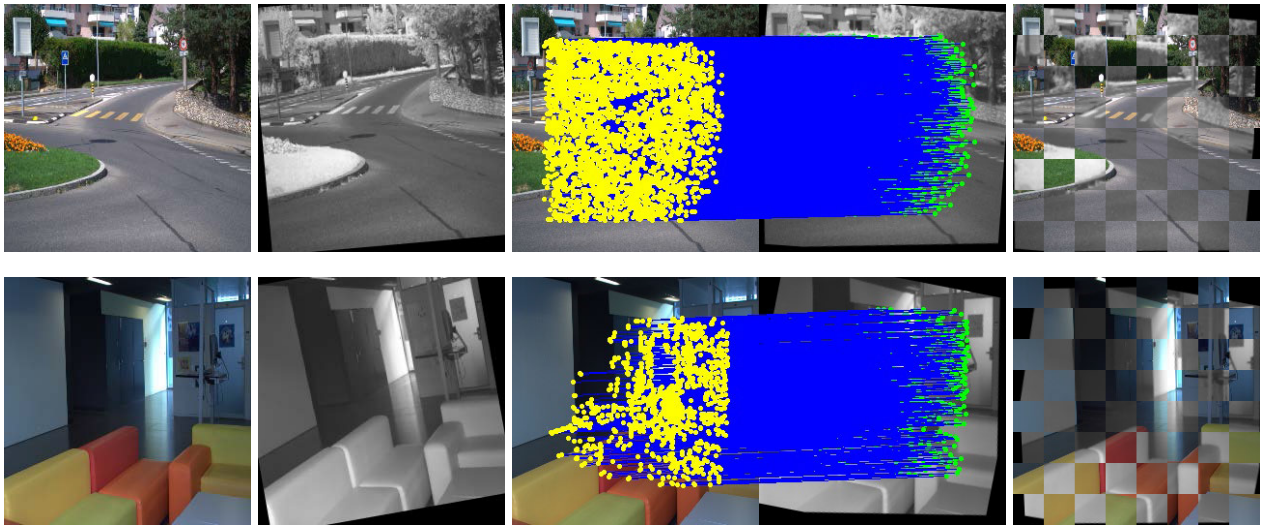


Fig. 12. Multimodal image registration results of the randomly selected images in the VIS-NIR dataset. (Left to right) VIS image, NIR image, matching point-pairs, and checkerboard image.

difficulty of image registration and improve the final image registration performance.

I. Visualizations

In this section, we show the visual registration results of multimodal images with different rotation angles. The proposed DOR-DDL can achieve many accurate correspondences between optical-SAR images in the SEN1-2 dataset (Fig. 10), VIS-NIR aerial images in the VEDAI dataset (Fig. 11), and VIS-NIR natural scene images in the VIS-NIR dataset (Fig. 12). The visualization checkerboard results show that the edges are very smooth, and there is no misalignment. These visual results demonstrate that DOR-DDL can register the multimodal images accurately under different degrees of RT.

J. Discussion of DOR-DDL

For the two-stage registration methods, the significant error in the first stage will influence the final registration performance. However, the proposed DOR network obtained a small rotation estimation error. In addition, the DDL network will solve the slight RT and achieve fine registration performances in the second stage. Thus, the introduced small rotation error will not significantly influence the final registration accuracy. The performance gains brought about by rotation correction are very significant at large RT. The proposed two-stage coarse-to-fine deep learning registration framework based on DOR and DDL networks is robust for multimodal image registration with RT. It should be noted that the coarse-to-fine deep learning registration method, DOR-DDL, is more suitable for multimodal images with large RT, and the single DDL network is sufficient for multimodal image registration without RT or small RT.

Moreover, when the maximum value of the rotation ordinal increases, the DOR network will design more binary classification branches for ordinal regression, and the network parameters of DOR will also increase. Correspondingly, the DOR network needs more training samples for optimization. In the future, we will design an iterative DOR network for

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rotation correction, which can predict the rotation iteratively without increasing the parameters of the DOR network.

V. Conclusion

This article proposed a novel coarse-to-fine deep learning registration framework for multimodal image registration. We provided an insightful analysis that the single DDL network is challenging to deal with the global geometric deformation caused by RT and significant local differences in appearance between multimodal images. Thus, this article decomposed the multimodal image registration into two stages: the rotation correction based on DOR and the fine image registration based on DDL. DOR can reduce the difficulty of the multimodal image registration task, significantly improving the performance of multimodal image registration with the RT. Moreover, this article transferred the rotation estimation task into an ordinal regression problem and proposed three effective DOR networks for rotation ordinal regression. Extensive experiments have shown the necessity of the two-stage coarse-to-fine deep learning registration method, the advantages of DOR in rotation correction, and the superiority of the proposed method in multimodal image registration. Noticeably, our proposal can bring obvious gains at a small time cost and computation cost. We will study fast and accurate multimodal image registration.

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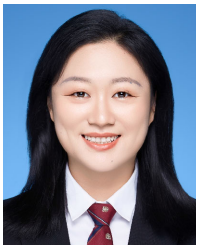
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